

Open Source Intelligence (OSINT) Notes

Open Source Intelligence (OSINT) is a critical component of modern intelligence gathering, cybersecurity, law enforcement, journalism, and corporate risk management. It involves collecting and analyzing publicly available data to produce actionable intelligence.

What is OSINT?

OSINT (Open Source Intelligence) refers to the process of gathering, analyzing, and using information that is available to the public. This includes data from the internet, social media, public records, news reports, and other openly accessible sources.

Importance of OSINT

1. Cybersecurity

- Detect threats, breaches, and leaked credentials.
- Monitor hacker forums and darknet marketplaces.
- Identify phishing domains and malicious IPs.

2. National Security & Law Enforcement

- Track terrorism, criminal activities, and smuggling.
- Map networks of threat actors and monitor movements.

3. Corporate Security

- Protect brand reputation.
- Identify internal leaks or fraud.
- Monitor competitors or mergers & acquisitions.

4. Journalism & Investigative Reporting

- Fact-checking.
- Exposing corruption and crimes using open-source investigations.

5. Humanitarian Aid & Crisis Response

- Monitor conflicts, disasters, and migration using real-time data.

OSINT Framework

An OSINT framework structures how information is collected and analyzed. Here's a basic overview:

1. Planning & Requirements

- Define the objective (e.g., "Identify who owns this suspicious domain").
- Determine key intelligence questions.

2. Collection

- Gather data from:
 - Websites
 - Social media
 - Public records
 - DNS and IP info
 - Leaked databases
 - Forums and blogs

3. Processing

- Clean, normalize, and format data.
- Use tools to extract metadata, timestamps, geo-location, etc.

4. Analysis

- Identify patterns, relationships, and anomalies.
- Build timelines, maps, and link charts.

5. Reporting

- Deliver actionable insights in a clear format (dashboard, document, briefing).

Common OSINT Tools & Techniques

Tools

Tool	Use Case
Maltego	Network mapping & link analysis
Shodan	Search for internet-connected devices
theHarvester	Email, domain, and metadata collection
SpiderFoot	Automated reconnaissance
Google Dorks	Advanced Google search techniques
Social-Searcher	Social media monitoring
FOCA	Document metadata extraction
Whois, DNSdumpster	Domain and IP lookup

Techniques

- **Google Dorking:** Advanced Google search queries to find sensitive info.
- **Reverse Image Search:** Identify image sources or manipulation.
- **Metadata Extraction:** Pull hidden info from files (e.g., author, location).
- **Geolocation:** Use visual clues to locate images or users.
- **Social Media Profiling:** Build a digital footprint of a person.
- **Domain Analysis:** Discover who owns and controls web infrastructure.
- **Dark Web Monitoring:** Watch underground forums and marketplaces.

Google Dorking

Google Dorking is a technique used to uncover hidden information using advanced search operators on Google. It can reveal sensitive data like passwords, login portals, unprotected documents, camera feeds, and misconfigured servers all indexed by Google but not meant to be public.

While it's a **legitimate tool for OSINT and cybersecurity auditing**, **malicious use of Google Dorking is illegal**.

What Is a Google Dork?

A **Google dork** is a specially crafted search query that uses **advanced operators** to find specific content.

Common Google Dorking Operators

Breakdown of the most important operators used in Google Dorking:

Operator	Description & Example
<code>site:</code>	Limits results to a specific domain.  <code>site:example.com</code>
<code>inurl:</code>	Finds URLs containing specific text.  <code>inurl:admin</code>
<code>intitle:</code>	Finds pages with keywords in the title.  <code>intitle:"index of"</code>

allintitle: All search terms must be in the title.

 **allintitle:login admin**

intext: Finds pages with keywords in the body text.

 **intext:"confidential"**

allintext: All words must appear in the text.

 **allintext:username password**

filetype: or
ext: Searches for specific file types.

 **filetype:pdf or ext:doc**

cache: Displays Google's cached version of a page.

 **cache:example.com**

link: Shows pages that link to a URL.

 **link:example.com**

related: Finds similar websites.

 **related:cnn.com**

define: Gets definitions.

 **define:OSINT**

* (wildcard)	Acts as a placeholder.
	 "admin * login"
OR	Finds results with either term.
	 login OR sign-in
-	Excludes a word.
	 filetype:pdf -site:gov
" (quotes)	Searches exact phrases.
	 "confidential report"

Sensitive Info You Can Find (Ethically, for security testing)

Type of Data	Example: Dork
Exposed documents	<code>filetype:xls</code> <code>site:example.com</code>
Admin login portals	<code>inurl:admin login</code>
Config/backup files	<code>intitle:index.of config</code>

Webcam streams	<code>inurl:view/view.shtml</code>
Passwords in code/files	<code>intext:"password="</code> <code>filetype:log</code>
Database dumps	<code>filetype:sql "password"</code>
Email lists	<code>filetype:xls</code> <code>intext:@gmail.com</code>

Legal & Ethical Warning

Google Dorking is a **powerful reconnaissance tool** used by:

- **Penetration testers**
- **Security researchers**
- **Cyber threat intelligence analysts**

However, **using it to exploit or access systems without permission is illegal** and can result in **criminal charges**.

Always follow **ethical hacking guidelines**, and use only on systems you have permission to test.

Example Use Case

Use Case: Tracking a cybercriminal

1. Start with an email from a phishing attempt.
2. Use [theHarvester](#) to find linked domains and social media.
3. Look up the IP address in [Shodan](#).
4. Find leaked databases containing that email.
5. Build a profile using social media and forum posts.
6. Generate a report of potential identities and locations.

Best Practices

- Always respect privacy and legal boundaries.
- Validate the accuracy of sources (OSINT ≠ truth).
- Use multi-source corroboration.
- Stay updated with new tools and techniques.

Comprehensive OSINT Tools List

1. People & Identity Search

Tool	Description
Pipl	Deep web search for people, emails, usernames
BeenVerified	Aggregates social media and public records

Spokeo	People search via phone, email, name
PeekYou	Profiles based on online activity
Social Searcher	Social media monitoring and search
Namechk	Checks username availability across platforms
That's Them	Free people search, phone, email, and IP lookups
EmailRep.io	Email reputation and breach status

2. Domain & Website Intelligence

Tool	Description
Whois Lookup	Domain registration details
Robtex	IP/domain relationships and routing
BuiltWith	Technology stack of websites
Netcraft	Site reports, hosting history

DNSdumpster	DNS mapping and subdomain discovery
crt.sh	Certificate transparency logs
VirusTotal	URL and file scanning for malware
Shodan	Search for internet-connected devices

3. Username & Account Tracking

Tool	Description
Sherlock	Finds usernames across many social platforms
Maigret	Similar to Sherlock, CLI tool for username check
WhatsMyName	Username check across platforms with detection
GHunt	Google account profiling via Gmail addresses

4. Geolocation & Mapping

Tool	Description
Google Earth/Maps	Satellite imagery, location mapping
EXIF.tools	Extracts image metadata for GPS/location info
GeoGuessr OSINT Edition	For geolocation training
SunCalc	Sun position calculator (used for photo analysis)
Mapillary	Street-level images around the world

5. Social Media Intelligence (SOCMINT)

Tool	Description
Twint	Twitter scraping without API
InVID	Verifies images and videos on social media
CrowdTangle	Facebook/Instagram content analysis

Snoopreport Instagram activity tracking (paid)

RedditSearch.io Reddit user search and post tracking

6. File & Metadata Analysis

Tool	Description
FOCA	Metadata extraction from docs (PDF, DOCX, etc.)
ExifTool	CLI tool for metadata extraction
DocDroid	View documents online without downloading

7. Automated Recon Frameworks

Tool	Description
SpiderFoot	Full-spectrum OSINT automation
Maltego	Graph-based link analysis tool
Recon-ng	Web recon framework in Python

OSINT Framework	Web-based directory of OSINT tools
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Sn0int	Semi-automated OSINT framework (CLI)
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8. Dark Web & Data Breach Search

Tool	Description
HavelBeenPwned	Check email for data breaches
DeHashed	Search leaked data (emails, usernames, hashes)
IntelligenceX	Dark web, breaches, archives
OnionSearchEngine	Search Tor hidden services

9. Infrastructure & Network Scanning

Tool	Description
Shodan	IoT and device search engine
Censys	Internet-wide scanning engine

ZoomEye Chinese version of Shodan

Nmap Port scanning and network mapping

10. Miscellaneous / Utilities

Tool	Description
Google Dorks	Use advanced Google search operators
Wayback Machine	View old versions of websites
Pastebin Scrapers	Monitor for leaked data
Tineye / Google Reverse Image Search	Image tracking
Hunter.io	Find email addresses associated with domains

OSINT Framework Websites

-  <https://osintframework.com/>